GATE

ALL BRANCHES

General Aptitude

QUANTITATIVE APTITUDE

Lecture No.- 05

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Recap of Previous Lecture

Topic Profit & Loss

$\frac{S.P.}{\boxed{C.P.}}$

Topics to be Covered

Topic-1 More on Profit Loss (Discount)

Topic-2 Mixtures & Alligation

[MCQ]

30%

#Q. Rohan spends 40% of his monthly income on food items and 50% of the remaining on clothes and conveyance. He saves one-third of the remaining amount after spending on food, clothes and conveyance. If he saves Rs. 19200 every year, what is his monthly income?

Assignment

(A) 32000

(B) 16000

(C) 12000

(D) 6000

$\frac{1}{3} \times \frac{0.5 \times 0.6}{0.1} \times x = 1600$

$\Rightarrow \frac{1}{3} \times 0.3 \times x = 1600$

$x = \frac{1600}{0.1} = 16000$

1600

$\frac{19200}{12}$

[MCQ]

#Q. 5% of income of P is equal to 15% of income of Q and 10% of income of Q equal 20% of income of R. If R's income is 2000, then What is total income of P, Q and R?

A. 9000

B. 12000

C. 15000

D. 18000

$5P = 15Q$

$\frac{P}{Q} = \frac{15}{5} = \frac{3}{1}$

$10Q = 20R$

$\frac{Q}{R} = \frac{20}{10} = \frac{2}{1}$

$R = 2000$

$Q = 4000$

$P = 12000$

18000

Assignment

$P : Q = 3 : 1$ ×2

$Q : R = 2 : 1$

$P : Q : R = 6 : 2 : 1$

2000

[MCQ]

#Q. A cloth merchant purchases cloth at ₹80/meter and sells at ₹100/meter. As a festive offer, he gives 50% extra free on every demand. Find his profit% or loss%.

A. 25% profit

B. 8.33% profit

C. 16.66% loss

D. 83.33% profit

Assignment

$\frac{100}{120} = \frac{5}{6}$

$0.1\overline{6}$

$16.\overline{6}\%$ loss

$\frac{S.P}{C.P} = \frac{100}{80} \times \frac{4}{6} = \frac{5}{6}$

$\frac{100}{80} \times \frac{5}{7.5}$

$\frac{1}{1.5}$

$\frac{10}{15}$

$\frac{2}{3}$

$= 0.8\overline{3}$

[MCQ]

$1 - 0.\overline{4} = 0.\overline{5}$

#Q. Due to downfall in the market, 'A' sells mangoes to 'B' saying "FOR EVERY 12 MANGOES, COUNT AS 8". Due to overnight demand, 'A' took his mangoes back from 'B' saying " FOR EVERY 8 MANGOES, COUNT AS 12". Find the profit or loss percent of 'A'.

$$\frac{S.P}{C.P} = \frac{\cancel{8}^{2}}{\cancel{12}_{3}} \times \frac{\cancel{8}^{2}}{\cancel{12}_{3}}$$

$$= \frac{4}{9} = 0.\overline{4}$$

$55.\overline{5}\%$ loss

PROFIT & LOSS

Note:

A trader may sometimes have multiple profits or losses simultaneously.

This is equivalent to having multiple changes and so all individual changes are to be multiplied to get the overall effect.

[MCQ]

$C.P = \frac{10}{15}$ $S.P = \frac{12}{12} = 1$

#Q. Sunidhi bought 15 apples for Rs.10 and sold them at the rate of 12 apples for Rs.12. What is the percentage of profit made by her?

each

$$\boxed{\frac{S.P}{C.P}} = 1 \div \frac{10}{15}$$

$$= 1 \times \frac{15}{10}$$

$$= 1.5$$

50% Profit

A. 100%

B. 150%

C. 125%

D. None of these

[MCQ]

'x' cm

#Q. A shopkeeper advertises for selling cloth at 4% loss. However by using a false meter scale he actually gains 25%. What is actual length of scale?

A. 76.8 cm

B. 77.8 cm

C. 74.8 cm

D. 75.8 cm

$$\frac{S.P}{C.P} = \boxed{0.96 \times \frac{100}{x} = 1.25}$$

$$\Rightarrow \frac{96}{1.25} = x$$

$$\therefore x = 76.8\text{ cm}$$

[MCQ]

#Q. A man sells an article at a profit of 20%. If he had bought it at 10% less and sold it for Rs. 18 more, he would have gained 40%. Find the cost price of the article.

$$\frac{S.P}{C.P} = 1.2 \Rightarrow S.P = 1.2\,C.P$$

$$\frac{S.P + 18}{0.9\,C.P} = 1.4$$

$$1.2\,C.P + 18 = 1.26\,C.P$$

$$0.06\,C.P = 18$$

$$C.P = \frac{18 \times 100}{0.06 \times 100} = \frac{1800}{6} = 300$$

Discounts:

$$\frac{S.P}{C.P}$$

[MCQ]

$S.P = 1.25\ C.P$ $\qquad \frac{S.P}{C.P} = 1.25$ $\qquad S.P_N = 60\%\ M.P$

#Q. By giving a discount of 25%, a shopkeeper gains 25%. If he gives a discount of 40%, find his gain or loss%.

No P No L

$S.P = 75\%\ M.P$

$S.P = 125\%\ C.P$

$75\%\ M.P = 125\%\ C.P$

$M.P = \frac{125}{75}\ C.P$

$M.P = \frac{5}{3}\ C.P$

M.P	C.P
Discount	Profit
25%	25%
40%	?

$60\%\ M.P = \frac{60}{100} \times \frac{5}{3}\ C.P$

$S.P_N = C.P$

$\frac{S.P_N}{C.P} = 1$

$$\frac{75}{C.P} = 1.25$$

$$\frac{75}{1.25} = C.P$$

60

M.P. = ₹ 100

S.P = ₹ 75

C.P = ₹ 60

S.P_N = ₹ 80

No P No L

[MCQ]

#Q. A trader gains 20% by giving a discount of 20%, if he gives a discount of 25% then find his P% or L%.

$\frac{9}{8} = 1.125$

.125

12.5% Profit

M.P = ₹100

S.P = ₹80

$\frac{80}{C.P} = 1.2$

$\frac{80}{1.2} = C.P = \frac{800}{12} = \frac{200}{3}$

Disc

20%

25%

Profit

20%

?

$S.P_M = 75$

$\frac{S.P_M}{C.P} = \frac{75}{} \times \frac{3}{200} = \frac{9}{8}$

$\times$ 75%

2 3

$80\% \, M.P = 100\% \, C.P$

$M.P = \frac{3}{2} C.P$

$\frac{S.P_N}{C.P} = \frac{9}{8} = 1.125$

12.5% Profit

[MCQ]

$S.P = 1.2\,C.P.$

#Q. An article was sold at a profit of 20%. If both cost price and selling price are ₹100 less each, then magnitude of the percentage of profit would have been 4 percentage points more than that in the first case. Then the cost price is

A. ₹ 500

B. ₹ 600

C. ₹ 800

D. None of these

$$\frac{S.P-100}{C.P-100} = 1.24$$

$$1.2\,C.P - 100 = 1.24\,C.P - 124$$

$$0.04\,C.P = 24$$

$$C.P = \frac{24}{0.04} = \frac{2400}{4}$$

600

[MCQ]

#Q. 5kg of ghee was bought by Vinod for ₹300. One kg from spoilt. He sells the remaining in such a way that on the whole he incurs a loss of 10%. At what price per kg does he sell the ghee?

Assignment

A ₹ 46.25

B ₹ 45.70

C ₹ 67.50

D ₹ 46.60

Puzzle:

$M + W + K = 100$

$(4M + 3W + \frac{1}{2}K = 100) \times 2$

#Q. A function hall was filled with 100 guests including men, women and kids. 100 biscuits has to be distributed among these guests, such that each man gets 4 biscuits, each woman as 3 and each kid gets 1/2 biscuits. How many men, women and kids are there in function hall?

$8M + 6W + K = 200$

$-\ M + W + K = 100$

$\boxed{7M + 5W = 100}$

Solution

2 280

A → 40/-

B → 1/-

C → 0.2

280

7 Men + 5 Women = 100

M → 5	×4	20	M – 10	40
W → 13	×3	39	W – 6	18
K → 82	×$\frac{1}{2}$	41	K – 84	42
100		100	100	100

Puzzle:

$A + B + C = 280$

$40A + B + 0.2C = 280$

TOTAL Money	:	280/-
TOTAL Number of Product	:	280

$A - 4$ 160

$B - 81$ 81

$C - 195$ 39

$\frac{A}{280}$

$\frac{A}{C} = \frac{8}{390} \quad \frac{4}{195}$

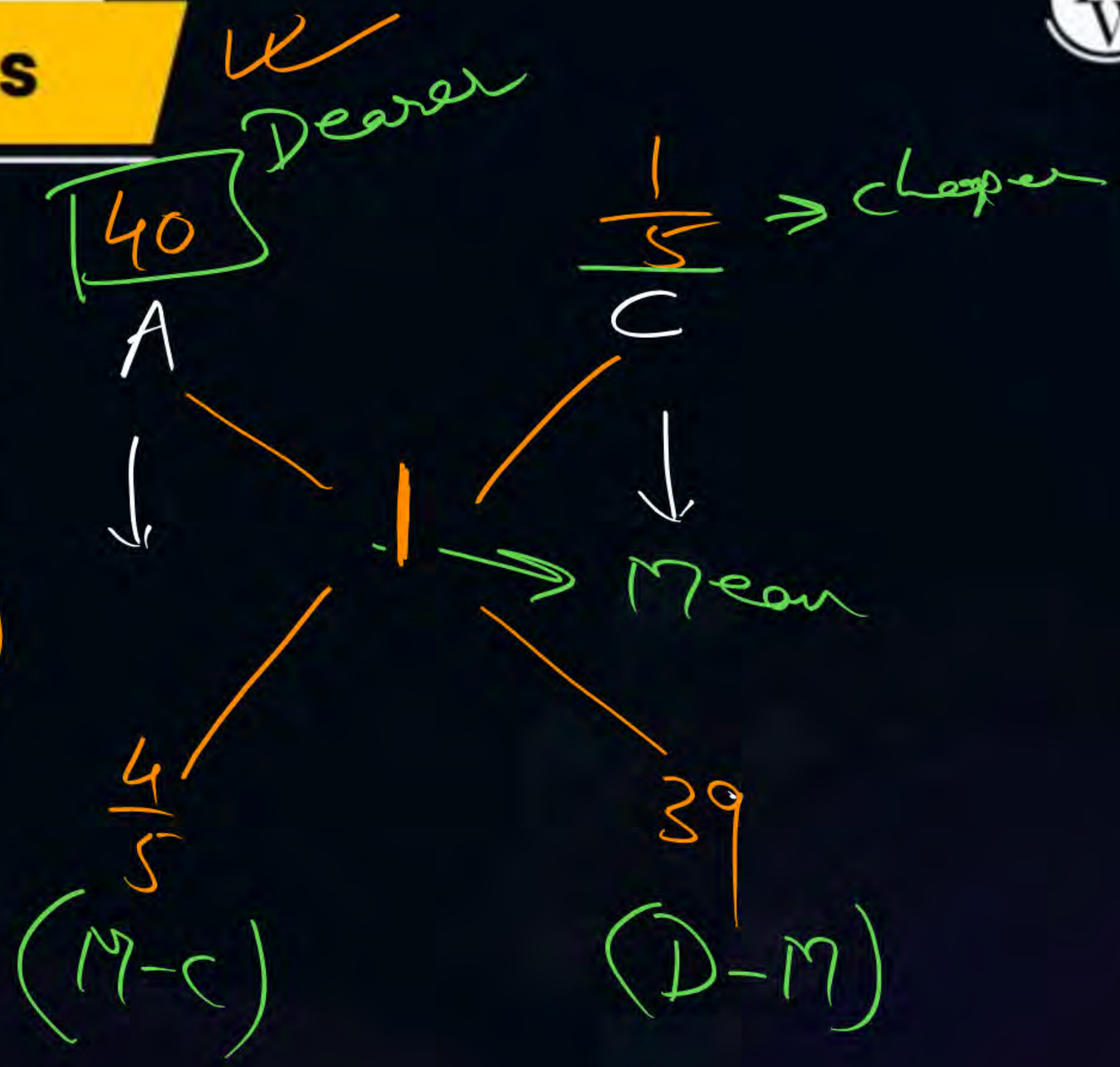
$A + \cancel{B} + C = 40A + \cancel{B} + 0.2C$

$$\boxed{39A = 0.8C}$$

Mixtures & Alligations

$A : C = \frac{4}{5} : 39$

$= 4 : 195$



[MCQ]

#Q. A trader purchases two varieties of rice 'A' & 'B' at the rate of ₹48/kg and ₹60/kg respectively. In what ratio he should mix he two varieties, so that the mixture cost becomes ₹52/kg?

A : B = 8 : 4

2 : 1

[MCQ]

#Q. A trader purchases two varieties of Sugar 'P' & 'Q' at the rate of ₹42/kg and ₹80/kg respectively. In what ratio he should mix he two varieties, so that he can sell the mixture at ₹78/kg by getting 30% profit?

S.P

S.P = 78

P% = 30%

$\frac{78}{C.P} = 1.3$

$\frac{78 \times 10}{13} = C.P = 60$

P : Q

= 20 : 18

10 : 9

[MCQ]

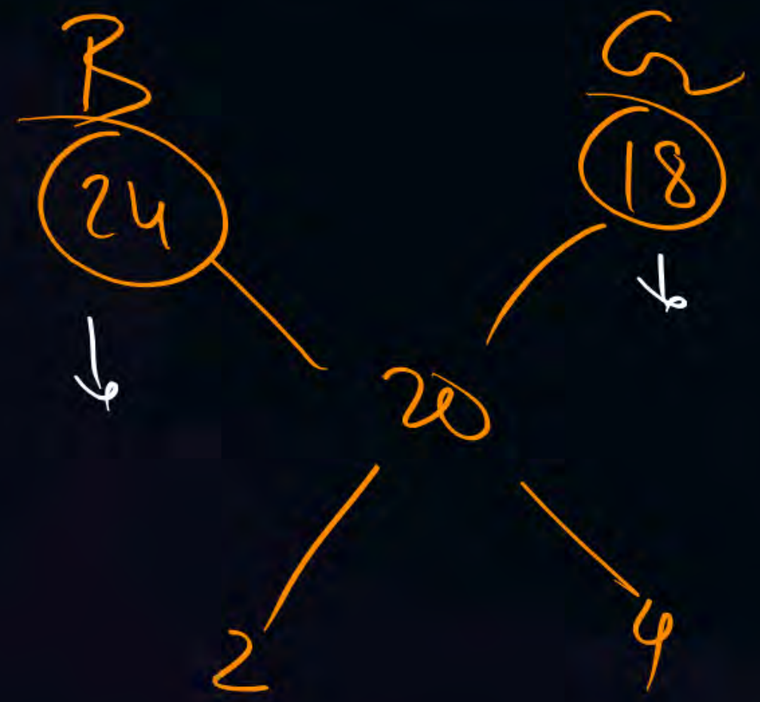
#Q. In a class with 360 students, the average age of all the boys is 24 years whereas average age of all the girls is 18 years. If average age of whole class is 20 years, then find the number of girls in that class.

A. 300

B. 40

C. 120

D. 240



$B : G = 2 : 4$

$= 1 : 2$

$\frac{2}{3} \times 360 = 240$

[MCQ]

#Q. A merchant sells 1600 horses at an overall profit of 25%. Some of them he sold at 20% profit whereas remaining at 40% profit. Find the number of horses he sold at 40% profit.

A. 400

B. 1200

C. 1240

D. 300

20% P : 40% P

$= \frac{15 : 5}{3 : 1}$

$\frac{1}{4} \times 1600 = 400$

20% 40%

25%

15 5

[MCQ]

#Q. In a mixture of 40 litres, the ratio of milk and water is 4:1. How much water must be added to this mixture so that the ratio of milk and water becomes 2:3?

$\frac{1}{5} \times 40$ $\frac{4}{5} \times 40$

M : W = 32 : 48

40 lit

Milk	Water
32 lit	8 lit

$\frac{32}{8+x} = \frac{2}{3}$

$\frac{32}{40+x} = \frac{2}{5}$

80 lit

W	M
48	32

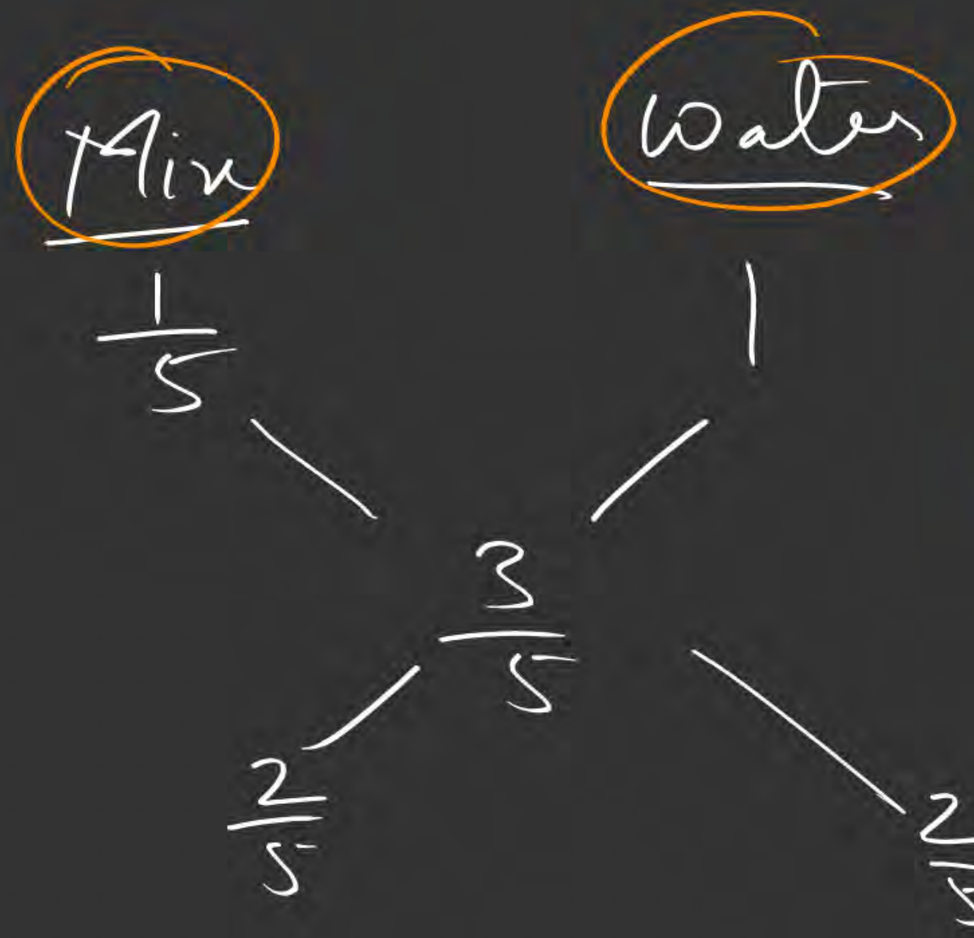
$\frac{8+x}{40+x} = \frac{3}{5}$

A. 20 litres

B. 30 litres

C. 32 litres

D. 40 litres



Mix : Water

$= \frac{2}{5} : \frac{3}{5}$

$= 1 : 1$

$= 40 : 40$

[MCQ]

#Q. An alloy A consist of iron & copper in the ratio 5 : 7, whereas alloy B consist iron & copper in the ratio 3 : 1. In what ratio these two alloys are to be mixed to form a new alloy with iron & copper in the ratio 2 : 1?

A 5:13

B 7:5

C 1:3

D 3:1

$A : B = \frac{1}{12} : \frac{3}{12}$

$= 1 : 3$

A

$\frac{5}{12}$

B

$\frac{3}{4} = \frac{9}{12}$

$\frac{8}{12} = \frac{2}{3}$

$\frac{1}{12}$

$\frac{3}{12}$

[MCQ]

#Q. How many litres of water must be added to 16 litres milk containing 10% water to dilute it with 20% water?

Assignment

A. 3 litres

B. 2 litres

C. 1 litre

D. 4 litres

[MCQ]

#Q. A mixture contains milk and water in the ratio 5:1. On adding 5 liters of water, the ratio of milk to water becomes 5:2. the quantity of milk in the mixture earlier was:

Assignment

A. 16 litres

B. 25 litres

C. 32.5 litres

D. 22.75 litres

[MCQ]

#Q. In what ratio must a grocer mix two varieties of pulses costing ₹15 and ₹20 per kg respectively so as to get a mixture worth ₹16.50 per kg?

Assignment

A. 3:8

B. 5:3

C. 7:3

D. 4:5

[MCQ]

#Q. In what ratio must Ankit mix two varieties of sugar worth ₹20 per kg and ₹32 per kg so that by selling the mixture at ₹36 per kg he may gain 20%?

Assignment

A. 1:5

B. 3:5

C. 2:3

D. 3:2

2 mins Summary

Topic Mixtures & Alligations

THANK - YOU